

Pull Request: Merge task-2 -> development

Summary

This PR completes Task 2: Quantitative Analysis using PyNance and TA-Lib for the Financial News Sentiment Analysis project. It includes comprehensive technical indicator calculations, financial metrics analysis, and visualization outputs for stock price data analysis.

Type of Change

- New feature (Quantitative analysis notebook)

Changes Made

Core Implementation

- Created notebooks/07_Quantitative_Analysis.ipynb: Comprehensive quantitative analysis notebook covering:

Technical Indicators Implemented

Moving Averages

- Simple Moving Average (SMA): Calculated for 20, 50, and 200-day periods

RSI (Relative Strength Index)

- 14-period RSI: Momentum oscillator (0-100 scale)

MACD (Moving Average Convergence Divergence)

- Standard parameters (12, 26, 9): MACD line, signal line, and histogram

Additional Indicators

- Bollinger Bands: Price volatility bands (20-period, 2 standard deviations)

Financial Metrics Calculated

Returns Analysis

- Daily Returns: Percentage change in closing prices

Risk Metrics

- Volatility: 30-day rolling volatility (annualized)

Additional Metrics

- Price Changes: Daily price changes and percentage changes

Visualizations Generated

Total of 5 high-resolution (300 DPI) visualization files saved to notebooks/figures/:

1. stock_prices_with_ma.png - Stock prices with moving averages for all 6 stocks

Dependencies

- Updated requirements.txt: Added pynance>=1.0.0 dependency

Key Findings

Technical Analysis Insights

1. Moving Averages: Successfully identify trends across multiple timeframes

Financial Metrics Insights

1. Returns: Comparative analysis shows performance differences across stocks

Trading Signals

- RSI overbought/oversold conditions identified for all stocks

Technical Details

Data Processing

- Loaded stock price data from CSV files for 6 stocks

Library Usage

- TA-Lib: Technical indicator calculations (if available)

Compatibility

- Graceful handling of missing TA-Lib library

Testing

- All calculations tested and verified

Documentation

- Comprehensive markdown documentation in notebook

Files Changed

- 1 new notebook file (07_Quantitative_Analysis.ipynb)

Performance Considerations

- Efficient data loading and processing

Security Considerations

- No sensitive data exposed

Next Steps After Merge

1. Proceed to Task 3: Correlation between News and Stock Movement

Related Work

- Task 1: Git and GitHub Setup & EDA (Completed)

Approval Checklist

- Code follows project structure
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Status: [DONE] Ready for merge to development

Impact: High - Establishes quantitative analysis foundation for correlation work

Risk: Low - Well-tested, documented, and structured implementation